v1.0Hadrons Deep Dive Request

V1.0 to V2.0 Versioning Delta — Tardigradia SubParticles Engine

Architecture: Hadronic Topological Architecture **Type:** Scientific Change Log | **Date:** June 2026
Applies to: `v1.0Hadrons Deep Dive Request V2.0.md`

Revision Necessity — Four Drivers

1. **Experimental discoveries 2022-2026** — LHC Run 3, LIGO O4, KATRIN 2024, NIF ignition, and other landmark results supersede V1.0 physics parameters
2. **New confirmed particles and phenomena** — Pines' Demon (2023), HH production evidence (2024), IceCube tau neutrino (2023), and others add entirely new simulation modes
3. **Platform upgrade** — WebGPU (Chrome 113+ 2023) replaces WebGL; enables 10^7+ particles
4. **Precision improvements** — CODATA 2022 constants, FLAG 2024 lattice QCD, NuFIT 5.3 oscillations

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V1.0 vs V2.0 Specific Changes

Parameter	V1.0	V2.0	Severity	Consequence if Not Updated
Exotic hadrons	Not in V1.0	LHCb 2022-2024: Tcc+, 3 pentaquarks confirmed. EXOTIC_HADRON type with up to 5 quarks	HIGH	Confirmed particles absent from simulation
QCD lattice masses	Experimental masses	BMW 2024: proton 938.3 MeV, neutron 939.6 MeV from first principles. Mass table replaced with lattice values	LOW	Slightly more accurate mass parameters
Nuclear sigma term	Not covered	FLAG 2024: sigma_piN = 38±4 MeV. Critical for dark matter-nucleon scattering	MEDIUM	Dark matter interaction rates depend on this parameter
String breaking fragmentation	V(r) = sigma*r only	Lund string: P_break = exp(-pi*m_q^2/sigma) per unit length. New quark pair creation visual	MEDIUM	Missing fundamental QCD mechanism for new particle production

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Simulation Impact Summary

The changes above drive the following modifications to the game engine architecture:

- **New particle types:** Exotic
- **Parameter updates:** All values in `static constexpr` declarations refreshed from 2022-2026 data
- **WebGPU migration:** Particle update loop moves from CPU JavaScript to WGSL compute shaders
- **New interaction modes:** Triggered by new experimental discoveries

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Unchanged Architecture

All foundational philosophy is **carried forward unchanged**: - Worldline Monism (One-Particle Universe) ontology - Proper time τ as primary particle identifier - Intrinsic vs. Extrinsic property distinction - Benevolence metric (C·F·S product formula) - Object-Oriented data structure (struct ParticleInstance extends to V2.0)

End of Versioning Delta | v1.0Hadrons Deep Dive Request V1.0 archived | V2.0 is the active specification as of June 2026